

Useful Imaginary Identities

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1 Definition

Define a number i such that

Definition 1.1.

$$\boxed{i^2 = -1}$$

2 Euler's equation

This formula is the underlying connection between imaginary numbers and the trigonometric functions.

Theorem 2.1. *Euler's equation*

$$\boxed{e^{ix} = \cos(x) + i \cdot \sin(x)} \quad (1)$$

Proof 1 through Taylor series expansion

Taking the Taylor series of $\cos(x)$, $\sin(x)$, e^{ix} we obtain

$$\begin{aligned} \cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \\ e^{ix} &= \sum_{k=0}^{\infty} \frac{(ix)^k}{k!} \end{aligned}$$

Note that we can split the Taylor series of e^{ix} into a real part and an imaginary part

$$\begin{aligned} e^{ix} &= \sum_{k=0}^{\infty} \frac{(ix)^k}{k!} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right) \\ e^{ix} &= \cos(x) + i \cdot \sin(x) \end{aligned}$$

Proof 2 through partial fraction decomposition

Evaluating the integral of $\int \frac{1}{x^2+1}$ two different ways

First way, change of variables

$$\begin{aligned} \text{Let } \tan(\theta) = x, \text{ then } dx &= \frac{1}{\cos^2(\theta)} d\theta \\ \int \frac{1}{x^2+1} dx &= \int \frac{1}{\frac{\sin^2(\theta)}{\cos^2(\theta)} + 1} \frac{1}{\cos^2(\theta)} d\theta = \int \frac{1}{\sin^2(\theta) + \cos^2(\theta)} d\theta = \int d\theta = \theta + C_1 \\ \int \frac{1}{x^2+1} dx &= \arctan(x) + C_1 \end{aligned}$$

Second way, partial fractions decomposition

$$\begin{aligned} \int \frac{1}{x^2+1} dx &= \int \frac{1}{(x+i)(x-i)} dx = \frac{i}{2} \cdot \int \frac{1}{x+i} dx - \frac{i}{2} \cdot \int \frac{1}{x-i} dx \\ \int \frac{1}{x^2+1} dx &= \frac{i}{2} \cdot \ln(x+i) - \frac{i}{2} \ln(x-i) + C_2 \\ \int \frac{1}{x^2+1} dx &= \frac{i}{2} \ln\left(\frac{x+i}{x-i}\right) + C_2 \end{aligned}$$

Equating the two expressions

$$\frac{i}{2} \ln\left(\frac{x+i}{x-i}\right) + C_2 = \arctan(x) + C_1$$

simplifying

$$\begin{aligned} -\ln\left(\frac{x-i}{x+i}\right) + C &= 2i \cdot \arctan(x) \\ \ln\left(\frac{x-i}{x+i}\right) + C &= 2i \cdot \arctan(x) \\ K \frac{x-i}{x+i} &= e^{2i \cdot \arctan(x)} \end{aligned}$$

Using a change of variables of $\tan(\theta) = x$

$$K \frac{\tan(\theta) - i}{\tan(\theta) + i} = e^{2i \cdot \theta}$$

simplifying

$$e^{2i \cdot \theta} = K \frac{(\tan(\theta) - i)^2}{\tan^2(\theta) + 1} = K \frac{\tan^2(\theta) - 2i \cdot \tan(\theta) - 1}{\frac{1}{\cos^2(\theta)}} = K(\sin^2(\theta) - i \cdot 2\sin(\theta)\cos(\theta) - \cos^2(\theta))$$

Using the double angle sin and cos formulas

$$e^{2i \cdot \theta} = K(-\cos(2\theta) - i\sin(2\theta))$$

Plugging in $\theta = 0$ we get

$$K = -1$$

Replacing 2θ with x

$$e^{i \cdot x} = \cos(x) + i \sin(x)$$

Proof 3 through differential equations

Consider the solutions of the differential equation $y' = i \cdot y$, with $y(0) = 1$

Notice that both $y(x) = e^{ix}$ and $y(x) = \cos(x) + i \sin(x)$ both satisfy the equation

Prove that the solution is unique

let $y_1(x), y_2(x)$ be two solutions to the above equation, then define

$$f(x) = y_1(x) - y_2(x)$$

$$\text{Notice that } f(0) = 1 - 1 = 0$$

$$f'(x) = y_1'(x) - y_2'(x) = i \cdot f(x)$$

To prove that f' is 0, hence f is constant, we define

$$g(x) = e^{-ix} f(x)$$

$$\text{Notice that } g(0) = 1 \cdot f(0) = 0$$

$$g'(x) = -i \cdot g(x) + e^{-ix} f'(x)$$

$$g'(x) = -i \cdot g(x) + i \cdot e^{-ix} f(x)$$

$$g'(x) = -i \cdot g(x) + i \cdot g(x) = 0$$

$$g(x) = 0 \quad \forall x$$

$$e^{-ix} f(x) = 0 \quad \forall x$$

Assuming e^{ix} is not 0 for any x

$$f(x) = 0 \quad \forall x$$

$$y_1(x) - y_2(x) = 0 \quad \forall x$$

$$e^{i \cdot x} = \cos(x) + i \sin(x) \quad \forall x$$

3 Properties

Starting from Euler's equation

Definition 3.1.

$$e^{ix} = \cos(x) + i \cdot \sin(x)$$

Using this equation we can plug in the value of $x = \frac{\pi}{2}$ and obtain

$$e^{i \frac{\pi}{2}} = \cos\left(\frac{\pi}{2}\right) + i \cdot \sin\left(\frac{\pi}{2}\right) = 0 + i \cdot 1 = i \quad (2)$$

Therefore we can claim

Definition 3.2.

$$i = e^{i \frac{\pi}{2}}$$

3.1 Reciprocal

$$\frac{1}{i} = \frac{1 \cdot i}{i \cdot i} = \frac{i}{-1} = -i \quad (3)$$

3.2 Square Root

$$\sqrt{i} = (e^{i\frac{\pi}{2}})^{\frac{1}{2}} = e^{i\frac{\pi}{4}} = \cos(\frac{\pi}{4}) + i \cdot \sin(\frac{\pi}{4}) = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \quad (4)$$

3.3 i to the power of i

$$i^i = (e^{i\frac{\pi}{2}})^i = e^{-\frac{\pi}{2}} \approx 0.208 \quad (5)$$